

Practical 5: Smoothing Techniques (Moving Average & SES)

Question: Given a monthly sales dataset, apply a 3-point and 5-point Moving Average. Then perform Simple Exponential Smoothing (SES) using $\alpha=0.3$. Compare the outputs and comment on how smoothing affects the series.

5.1 Objective

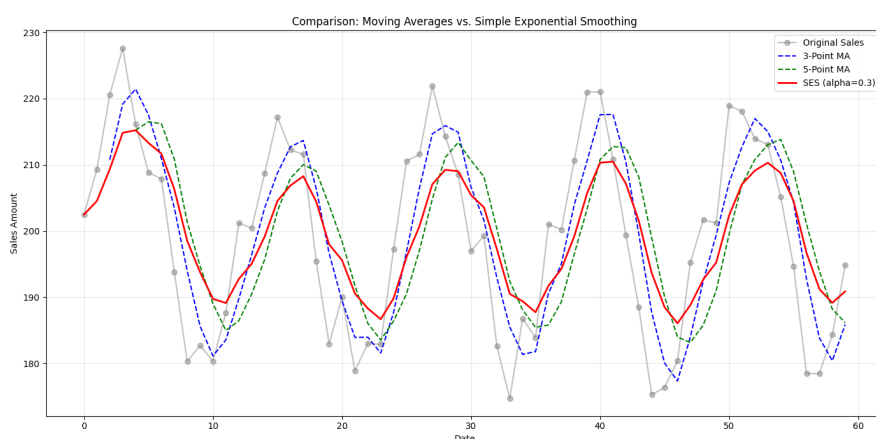
- To apply Moving Average (MA) and Simple Exponential Smoothing (SES) to a time series.
- To compare the smoothing effects of different window sizes and smoothing parameters.

5.2 Theory

- **Moving Average (MA):** Calculates the average of data points within a sliding window. Larger windows produce smoother curves but introduce more lag.
- **Simple Exponential Smoothing (SES):** Assigns exponentially decreasing weights to past observations. The parameter α ($0 < \alpha < 1$) controls the weight given to the most recent observation.

5.3 Code Implementation

```
# 1. Moving Averages # Calculate 3-point and 5-point rolling means df['MA_3'] =  
df['Sales'].rolling(window=3).mean() df['MA_5'] = df['Sales'].rolling(window=5).mean()  
# 2. Simple Exponential Smoothing (SES)  
# Apply SES with alpha = 0.3 df['SES_0.3'] = df['Sales'].ewm(alpha=0.3,  
adjust=False).mean()
```



5.4 Interpretation

- **MA Comparison:** The 5-point MA is smoother than the 3-point MA but reacts slower to the upward trend (more lag).
- **SES vs MA:** The SES ($\alpha=0.3$) provides a balanced smooth curve that tracks the original data's trend closer than the 5-point MA, making it more responsive to recent changes.